

Progress Report

Real Estate Pricing Intelligence Dashboard

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Project Type: Executive Dashboard / Interactive Analytics System

Dashboard Purpose: This project is designed to help buyers, sellers, real estate agents, investors, and analysts quickly understand whether properties are selling above, below, or near their estimated value. The dashboard answers a practical question: Is a property fairly priced before making a buying, selling, or investment decision?

Purpose of this Report: This progress report documents the current state of development of my real estate analytics dashboard, including completed work, data preparation, visualizations, interactivity, challenges, changes from the proposal, remaining work, and questions for instructor feedback.

Current Prototype Status:

- Cleaned dataset prepared and ready for visualization
- Five working charts connected to real data
- Interactive filters implemented and functioning
- Dashboard updated to use number of bedrooms instead of property type
- Analytical pipeline working from CSV upload to visualization
- Remaining work focused on refinement, testing, and final presentation polish

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1. Introduction

This report documents the progress made in developing an interactive real estate analytics dashboard. The goal of this project is to analyze discrepancies between estimated property values and actual sale prices using cleaned data, derived metrics, and interactive visualizations.

The dashboard is now positioned as a real estate pricing intelligence tool. It is intended for users who need to quickly understand whether properties are overpriced, underpriced, or fairly valued. Instead of requiring users to interpret raw transaction data manually, the dashboard summarizes pricing behavior through KPI cards, visual comparisons, trend analysis, and interactive filtering.

A key design improvement made during the prototype phase was making the dashboard purpose visible immediately to the user. The dashboard now clearly states that it is built for buyers, sellers, agents, and investors, and that its main purpose is to compare actual sale prices with estimated values before a decision is made.

This is important because inaccurate pricing can directly impact buying decisions, investment returns, and market efficiency.

1.1 Data Source

[Click here for dataset source](#)

Note: In addition to the dashboard implementation, Python scripts were used for data validation, preprocessing verification, and visualization testing, and have been included in the GitHub repository for reproducibility.

Project Repository: [Click here for GitHub Repository Link](#)

The dataset used in this project was obtained from Kaggle and consists of real estate transaction records, including sale price, estimated value, room count, locality, date-related fields, and other property attributes. The dataset was selected because it directly supports valuation analysis and contains enough records to create meaningful patterns across time, location, valuation category, and property size.

One important change from the earlier version of the project is that the dashboard no longer uses property type labels such as “Single Family,” “Two Family,” or “Four Family” as the main segmentation field. Based on instructor feedback, the dashboard now uses number of bedrooms instead because it is easier for users to understand and more directly connected to property size.

2. 1. What I Have Completed

At this stage of the project, I have developed a functional working prototype of a real estate analytics dashboard. The system integrates a complete data pipeline, multiple visualizations, dynamic

filtering, and user-facing explanations. All implemented components operate on real cleaned data and produce meaningful analytical outputs.

2.1 1.1 Data Pipeline Implementation

A complete end-to-end data pipeline has been implemented to transform raw CSV data into structured, analysis-ready data suitable for dashboard visualization.

The pipeline consists of the following stages:

1. **Data Ingestion:** The dataset is uploaded through a file input interface and parsed using a CSV parser. This allows the dashboard to load the cleaned real estate dataset directly and produce visualizations dynamically.
2. **Data Cleaning:** The raw dataset required cleaning before it could be used for analysis. The cleaning process addressed missing values, invalid entries, non-numeric values in numeric fields, and inconsistent category formatting. Cleaning was necessary because charts using raw data could produce misleading results or fail entirely.
3. **Type Conversion:** Important numeric fields such as sale price, estimated value, price difference, percentage difference, number of rooms, number of bathrooms, and carpet area were converted into numeric format. This ensures that mathematical calculations and chart aggregations are accurate.
4. **Feature Engineering:** New analytical fields were used or created to support interpretation:
 - Price Difference = Sale Price – Estimated Value
 - Percentage Difference
 - Valuation Category (Overvalued / Fair / Undervalued)
 - Number of Bedrooms, derived from available room or bedroom-related fields
5. **Temporal Aggregation:** Date fields were transformed into monthly time periods to support time-series analysis. A 3-month moving average was applied to reduce noise and reveal clearer pricing trends over time.
6. **Dashboard Transformation:** The cleaned and transformed data is passed into React dashboard components, where it is filtered, aggregated, and visualized in real time.

This pipeline ensures that all dashboard outputs are based on structured, validated, and consistent data.

2.2 1.2 Data Cleaning Logic Before and After

The cleaning process was critical to ensuring accuracy and usability.

Before Cleaning:

- Missing values existed in important fields such as sale price and estimated value
- Some numeric columns required conversion before calculations could be performed
- Some categorical fields contained unclear or unusable values
- Property type labels were not ideal for the final dashboard because they were less intuitive for users
- Some records included values outside the main market range and created visual crowding

After Cleaning:

- Required numeric fields are converted and valid
- Missing or invalid critical values are removed or handled
- Locality and valuation category fields are standardized
- Number of bedrooms is used as a clearer segmentation field
- Derived metrics are available for analysis
- Charts are filtered to a readable \$100K–\$2M range when appropriate

The cleaning process ensures that downstream calculations and visualizations are reliable. It also improves the user experience by preventing confusing labels, invalid outputs, and chart clutter.

2.3 1.3 Valuation Classification Logic

Each property is classified by comparing actual sale price to estimated value. The main formula is:

$$\text{Percentage Difference} = \frac{\text{Sale Price} - \text{Estimated Value}}{\text{Estimated Value}} \times 100$$

The dashboard uses the following classification system:

- **Overvalued:** Sale price is more than 10% above estimated value
- **Fairly Priced:** Sale price is within $\pm 10\%$ of estimated value
- **Undervalued:** Sale price is more than 10% below estimated value

The $\pm 10\%$ threshold was chosen to allow for normal market variation while still identifying meaningful pricing deviations. This methodology is displayed visually in the dashboard so users can understand the classification before viewing the KPI cards and charts.

2.4 1.4 Implemented Visualizations

Five core visualizations have been successfully implemented. All use real data and dynamically respond to user interaction.

2.4.1 1. KPI Summary Dashboard

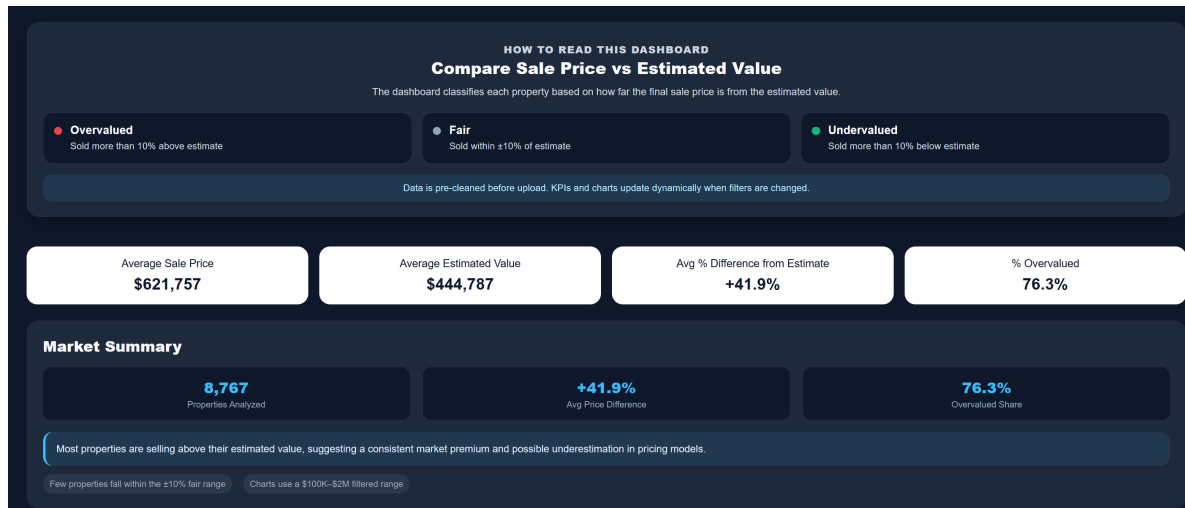


Figure 1: High-level metrics summarizing overall pricing trends before detailed analysis.

Description: The KPI summary cards display high-level metrics:

- Average Sale Price
- Average Estimated Value
- Average Percentage Difference from Estimate
- Percentage of Overvalued Properties

Insight: The average sale price consistently exceeds the average estimated value, indicating a systematic underestimation of market demand and suggesting that estimation models may lag behind real transaction behavior.

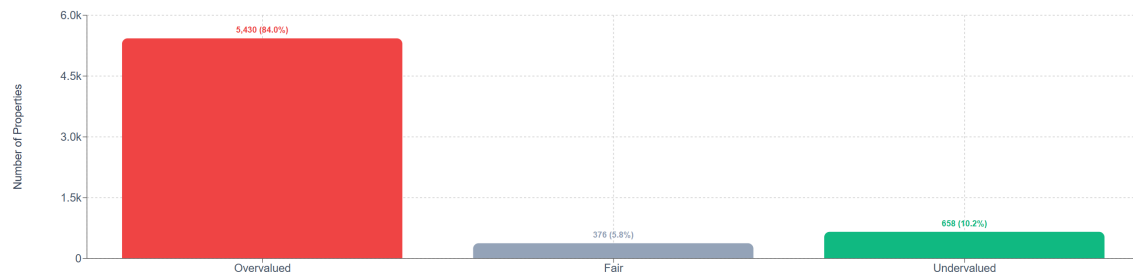
Purpose: The KPI cards provide an immediate overview before the user explores individual charts. This helps users quickly understand the market direction without reading a long explanation.

UI/UX Improvement: A separate “How to Read This Dashboard” section was added before the KPI cards so users understand what Overvalued, Fair, and Undervalued mean before interpreting the summary metrics.

2.4.2 2. Valuation Category Distribution

Properties by Valuation Category

This chart summarizes how many properties are overvalued, fairly priced, or undervalued within the \$100K to \$2M range.



Insight: Over 75% of properties fall into the overvalued category, indicating a strong upward pricing bias relative to estimated values. Fairly priced properties represent only a small portion of the dataset.

Figure 2: Distribution of properties across overvalued, fair, and undervalued categories.

Description: This bar chart shows the number of properties categorized as:

- Overvalued
- Fairly Priced
- Undervalued

Insight: A large majority of properties fall into the overvalued category, suggesting that many transactions exceeded estimated values.

Purpose: This chart quantifies the overall imbalance in valuation accuracy and gives a simple count-based summary of the market.

2.4.3 3. Scatter Plot: Sale Price vs Estimated Value



Figure 3: Comparison of sale price and estimated value showing overvaluation and undervaluation relative to the fair value line.

Description: This scatter plot compares each property's estimated value with its actual sale price. A dashed fair value line is included where sale price equals estimated value.

Insight: Most points lie above the fair value line, reinforcing that sale prices are generally higher than estimated values. The spread also widens at higher price levels, suggesting that pricing inefficiencies may increase for higher-value properties.

Purpose: The scatter plot allows visual identification of deviation patterns, outliers, and whether the market generally sells above or below estimates.

Dashboard Update: The tooltip was updated to remove property type labels and show number of bedrooms instead. This makes the chart consistent with the revised dashboard design and instructor feedback.

2.4.4 4. Monthly Pricing Trend

Monthly Pricing Trend

This chart compares 3-month moving average trends of sale price and estimated value over time, aggregated at the monthly level.



Insight: Sale prices stay above estimated values across the majority of the timeline, with the gap widening during 2014–2016 and again after 2019. This suggests sustained market pressure and possible underestimation in pricing models, alongside cyclical fluctuations that may reflect seasonal or macroeconomic effects.

Figure 4: 3-month moving average trend showing sale prices consistently exceeding estimated values over time.

Description: This line chart displays a 3-month moving average of:

- Sale Price
- Estimated Value

Insight: The gap between sale price and estimated value increases during certain periods, especially around the mid-2010s and after 2019. This may indicate stronger market demand or delayed adjustment in estimation models.

Extended Insight: This pattern suggests that the estimation model may not adapt quickly to changing market conditions. During rapid growth periods, sale prices may rise faster than estimated values, creating a persistent gap.

Purpose: This chart reveals long-term trends and cyclical behavior that may not be visible in static summaries.

2.4.5 5. Bedroom-Based Pricing Comparison

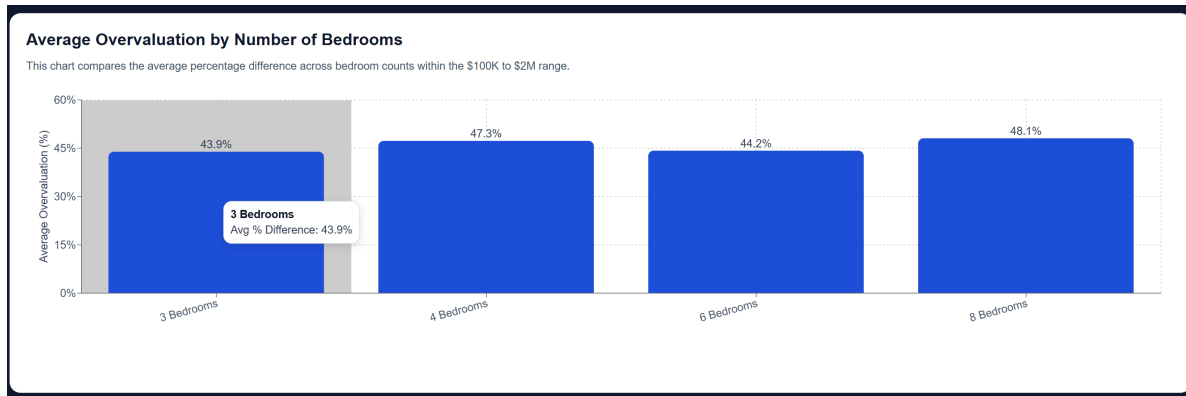


Figure 5: Average pricing difference segmented by number of bedrooms to reflect property size more intuitively.

Description: This bar chart compares average percentage difference by number of bedrooms instead of property type. The previous version grouped properties by labels such as Single Family or Two Family, but this was changed because bedroom count is easier for users to interpret.

Insight: Bedroom count provides a clearer and more intuitive segmentation of property size. Initial analysis suggests that higher-bedroom properties may exhibit larger deviations, indicating that estimation accuracy may decrease as property size increases.

Purpose: This chart provides segmentation-level insight based on property size using a more intuitive and user-friendly metric.

2.5 1.5 Interactive Features

The dashboard includes fully functional interactivity, allowing users to explore the dataset dynamically.

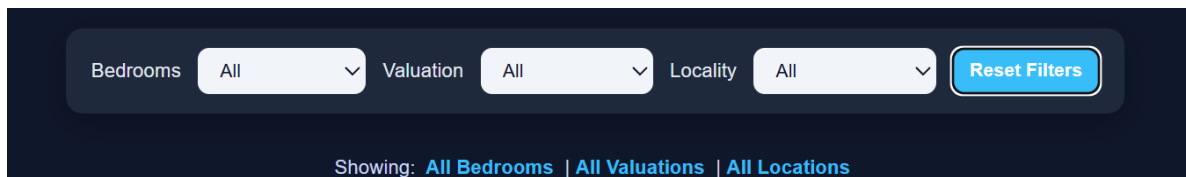


Figure 6: Interactive filters enabling dynamic analysis by bedrooms, valuation category, and locality.

Implemented Filters:

- Number of Bedrooms Filter
- Valuation Category Filter

- Locality Filter

Behavior:

- All filters are applied globally
- KPI cards update based on selected filters
- All charts update simultaneously based on selected filters
- The active filter summary shows the current dashboard state
- The reset button returns all filters to the default view

Core interactivity is fully functional:

- Bedroom filter (e.g., 1 Bedroom, 2 Bedrooms, 3 Bedrooms)
- Valuation Category filter (Overvalued, Fair, Undervalued)
- Location filter using locality

These filters dynamically update all charts in real time, ensuring consistent cross-visualization analysis. This ensures that the dashboard is not static but supports exploratory analysis.

2.6 1.6 UI/UX Improvements Completed

Based on instructor feedback, several interface improvements were made.

- The dashboard title was changed to **Real Estate Pricing Intelligence** to make the purpose clearer.
- A tagline was added: users can immediately understand that the dashboard helps identify overpriced, underpriced, and fairly valued properties.
- The target audience is shown at the top: buyers, sellers, agents, and investors.
- The long methodology paragraph was redesigned into a visual “How to Read This Dashboard” card.
- KPI insights were converted from long text into a more scannable market summary section.
- Property Type was replaced with Bedrooms to make the interface more understandable to general users.
- Tooltip labels were updated to show bedrooms instead of property type.

These changes directly address the concern that users should understand the dashboard without needing a verbal explanation.

These improvements ensure that users can understand the dashboard purpose and insights within seconds, without relying on detailed reading.

2.7 1.7 System Functionality Status

The current prototype meets the definition of a working system:

- All visualizations render correctly
- All charts use real cleaned data
- Filters dynamically update outputs
- The data pipeline runs from CSV upload to visualization without errors
- Empty or unloaded states are handled more clearly
- The dashboard is now easier for non-technical users to understand

While the interface can still be refined, the system is fully operational and analytically meaningful.

3. 2. Challenges and Solutions

3.1 2.1 Data Quality and Cleaning Challenges

Problem: The dataset required cleaning before it could be visualized. Numeric values needed conversion, some fields contained missing values, and certain categories were not user-friendly.

What I Tried: I initially attempted to use the available property type field for filtering and segmentation. However, labels such as Single Family, Two Family, and Four Family raised usability concerns because they may not be immediately meaningful to every user.

Solution: The dashboard was revised to use number of bedrooms as the main property-size filter. This better matches how users commonly think about real estate and creates a clearer interface.

3.2 2.2 Visualization Clutter

Problem: The original scatter plot used too many points and appeared crowded. It was difficult to interpret because the lower price range contained many overlapping observations.

What I Tried: I experimented with log scaling, filtering, point opacity, and limiting the visible market range.

Solution: The final scatter plot uses a \$100K–\$2M range, log-scaled spacing, point sampling for readability, and a fair value reference line. This makes the chart easier to interpret while preserving the important pricing pattern.

3.3 2.3 Dashboard Purpose Was Not Immediately Clear

Problem: Instructor feedback identified that a user should not have to ask what the dashboard is for. The dashboard needed to explain its purpose visually and quickly.

What I Tried: The first version used a simple title and subtitle, but it was not strong enough to communicate user value.

Solution: The top section was redesigned with a clear audience label and value statement:

- For Buyers, Sellers, Agents, and Investors
- See if properties are overpriced, underpriced, or fairly valued before making a decision

This makes the dashboard's purpose visible at a glance.

3.4 2.4 Long Text Reduced Readability

Problem: The methodology and insight sections originally used long paragraphs. Instructor feedback noted that users often do not read long blocks of text in dashboards.

Solution: The methodology section was redesigned into a visual card explaining Overvalued, Fair, and Undervalued categories. The market summary section was also redesigned to show compact KPI-style insights instead of long paragraphs.

4. 3. Changes from Proposal

Several changes were made from the original project direction:

- One of the most significant changes from the original proposal was replacing the Property Type variable (e.g., Single Family, Two Family) with Number of Bedrooms as the primary segmentation feature. While the original plan relied on property type categories, it became clear during development and instructor feedback that these labels were less intuitive for general users. Bedroom count provides a clearer and more universally understood measure of property size. This change improved both usability and interpretability of the dashboard, making it easier for users to quickly analyze pricing patterns.
- The dashboard became more user-centered, with a stronger emphasis on buyers, sellers, agents, and investors.
- Property Type filtering was replaced with Number of Bedrooms filtering.
- The property type comparison chart was replaced with a bedroom-based pricing comparison chart.
- The dashboard title and tagline were rewritten to better explain purpose.
- The methodology section was redesigned from text-heavy explanation into a scannable visual guide.
- The market summary section was redesigned to make insights easier to read.
- The scatter plot tooltip was updated to show bedroom count instead of property type.

These changes improved usability, reduced confusion, and made the dashboard more understandable for a general audience.

5. 4. Remaining Work

The current prototype is functional, but several refinements remain for the final submission:

- Continue testing all filter combinations
- Verify that all charts update correctly after filtering by bedrooms, valuation category, and locality
- Add final screenshots to the written report
- Improve responsive design for smaller screen sizes
- Consider adding click-to-filter interactivity as a final enhancement
- Review all chart captions and report text for consistency with the updated dashboard
- Prepare the final presentation and demo video

The most important remaining priority is stability. The dashboard must continue to work correctly across normal and edge-case filter selections.

6. 5. Questions for Instructor

- Should the dashboard prioritize additional interactivity, such as click-to-filter, or focus more on visual polish?
- Are the current explanations short enough for dashboard users, or should they be reduced even further?
- Is the \$100K–\$2M filtered range appropriate for readability, or should the dashboard offer a visible price range control?

7. Conclusion

This project demonstrates the ability to transform raw real estate transaction data into meaningful insights through cleaning, analysis, visualization, and user-centered design. The current dashboard is no longer only a technical visualization project; it is now positioned as a practical real estate pricing intelligence tool.

The most important progress made so far includes building a complete data pipeline, implementing five working visualizations, adding global filters, improving the dashboard's purpose statement, redesigning methodology explanations, and replacing property type labels with bedroom-based

analysis. These changes make the dashboard easier to understand and more useful for buyers, sellers, agents, investors, and analysts.

More importantly, the project shows the importance of questioning design choices, validating assumptions, and making analytical tools understandable to users without requiring extra verbal explanation.

Overall, the project reflects a shift from simply building charts to designing a user-focused analytical tool that communicates insights clearly and efficiently.